to_enum → Enumerator
: Converts an object (presumably a collection) to an enumerator. It's never called internally by the interpreter.

to_hash → Hash
: Used when the interpreter expects to see Hash, as in when converting a ** in a method call.

to_int → Integer
: Used when the interpreter expects to see an integer value (such as a file descriptor or as a parameter to Integer).

to_io → IO
: Used when the interpreter is expecting I/O objects (for example, as parameters to the methods IO.reopen or IO.select).

to_open → IO
: Called (if defined) on the first parameter to IO.open.

to_path → String
: Called by the interpreter when it is looking for a filename (for example, by File.open)

to_proc → Proc
: Used to convert an object prefixed with an ampersand in a method call.

```
class OneTwo
  def to_proc
    proc { "one-two" }
  end
end
def silly
  yield
end

ot = OneTwo.new
silly(&ot) # => "one-two"
```

to_regexp → Regexp
: Invoked by Regexp#try_convert to convert its argument to a regular expression.

to_str → String
: Used pretty much any place the interpreter is looking for a String value. Except for string interpolation, which uses to_s.

```
class OneTwo
  def to_str
    "one-two"
  end
end

ot = OneTwo.new
puts("count: " + ot)
File.open(ot) rescue puts $!.message
```

produces:

```
count: one-two
No such file or directory @ rb_sysopen - one-two
```